

Problem

Repeat the previous problem if $R_f = 10\text{ k}\Omega$.

Step-by-step solution

Step 1 of 4

Draw the circuit diagram as shown in Figure 1.

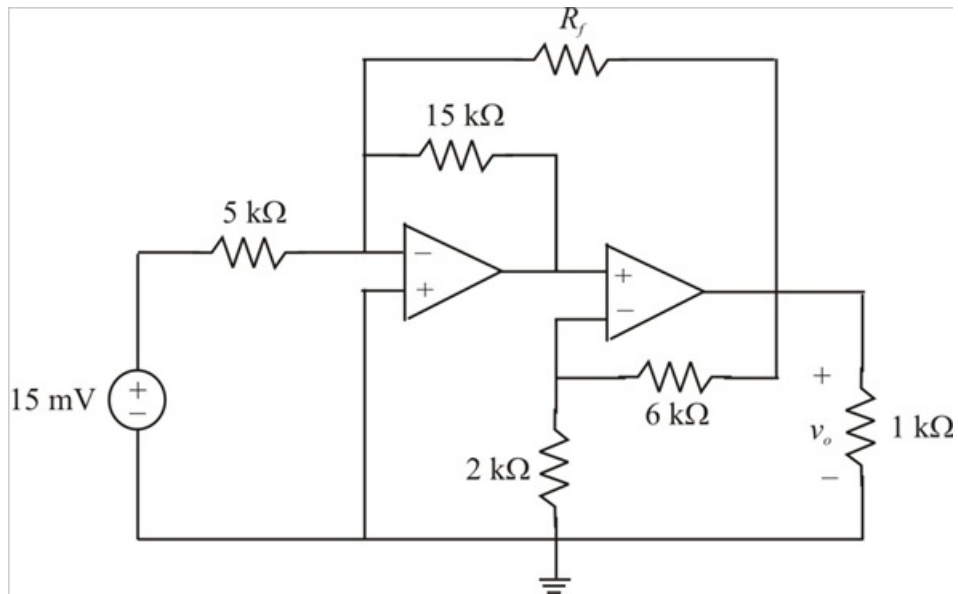


Figure 1

Step 2 of 4

The circuit is labeled as shown in Figure 2.

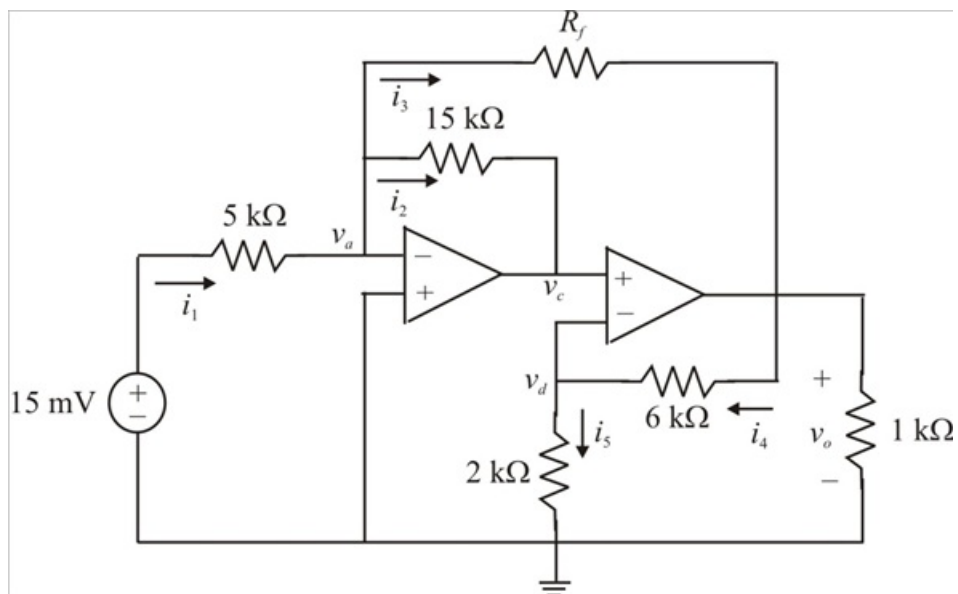


Figure 2

Step 3 of 4

The feedback resistance $R_f = 10\text{ k}\Omega$

Applying Kirchhoff's current law to node v_a ,

$$i_1 = i_2 + i_3$$

$$\frac{(15 \times 10^{-3}) - v_a}{5} = \frac{v_a - v_c}{15} + \frac{v_a - v_o}{10}$$

$$(90 \times 10^{-3}) - 6v_a = 2v_a - 2v_c + 3v_a - 3v_o$$

$$11v_a - 2v_c - 3v_o = 90 \times 10^{-3}$$

Therefore $11v_a - 2v_c - 3v_o = 90 \times 10^{-3}$ (1)

But $v_a = 0$ since non-inverting terminal is grounded. Then equation (1) becomes,

$$11v_a - 2v_c - 3v_o = 90 \times 10^{-3}$$

$$11(0) - 2v_c - 3v_o = 90 \times 10^{-3}$$

$$-2v_c - 3v_o = 90\text{ mV}$$

Therefore $-2v_c - 3v_o = 90\text{ mV}$ (2)

Step 4 of 4

Applying Kirchhoff's current law to node v_d ,

$$i_4 = i_5$$

$$\frac{v_o - v_d}{6} = \frac{v_d - 0}{2}$$

$$v_o - v_d = 3v_d$$

$$v_d = \frac{v_o}{4}$$

Therefore $v_d = \frac{v_o}{4}$ (3)

But $v_c = v_d$ for an ideal op amp, substituting in equation (2), we get

$$-2v_c - 3v_o = 90\text{ mV}$$

$$-2v_d - 3v_o = 90 \times 10^{-3}$$

$$-2\left(\frac{v_o}{4}\right) - 3v_o = 90 \times 10^{-3}$$

$$-14v_o = 360 \times 10^{-3}$$

$$v_o = -25.71\text{ V}$$

Therefore the output voltage v_o is -25.71 mV .